

Inequalities - Solutions

Problem 1. Find the greatest real number C such that

$$\sqrt{\frac{x}{y+z}} + \sqrt{\frac{y}{z+x}} + \sqrt{\frac{z}{x+y}} > C \quad (1)$$

for any positive real numbers x, y, z .

Solution 1. The answer is $C = 2$. Consider $x = y$ and $z \rightarrow 0$. Then the left-hand side of the inequality tends to $1 + 1 + 0 = 2$. So C cannot exceed 2.

It remains to prove (1) for $C = 2$. Since the inequality is homogeneous, we may assume $x + y + z = 1$. The left-hand side of (1) becomes

$$\sqrt{\frac{x}{1-x}} + \sqrt{\frac{y}{1-y}} + \sqrt{\frac{z}{1-z}}.$$

We claim that the first term is at least $2x$. Indeed,

$$\begin{aligned} \sqrt{\frac{x}{1-x}} &\geq 2x \\ \Leftrightarrow x &\geq 4x^2(1-x) \\ \Leftrightarrow x(2x-1)^2 &\geq 0. \end{aligned}$$

This clearly holds. It follows that

$$\sqrt{\frac{x}{1-x}} + \sqrt{\frac{y}{1-y}} + \sqrt{\frac{z}{1-z}} \geq 2x + 2y + 2z = 2.$$

Equality holds when $x = y = z = \frac{1}{2}$, which violates the condition $x + y + z = 1$. So there is no equality case. This proves (1).

Solution 2. This gives an alternative proof for

$$\sqrt{\frac{x}{y+z}} + \sqrt{\frac{y}{z+x}} + \sqrt{\frac{z}{x+y}} > 2.$$

By Hölder's inequality, we have

$$(x^2(y+z) + y^2(z+x) + z^2(x+y)) \left(\sqrt{\frac{x}{y+z}} + \sqrt{\frac{y}{z+x}} + \sqrt{\frac{z}{x+y}} \right)^2 \geq (x+y+z)^3.$$

Rearranging the terms, this yields

$$\left(\sqrt{\frac{x}{y+z}} + \sqrt{\frac{y}{z+x}} + \sqrt{\frac{z}{x+y}} \right)^2 \geq \frac{(x+y+z)^3}{x^2y + x^2z + y^2x + y^2z + z^2x + z^2y}.$$

It remains to prove

$$\frac{(x+y+z)^3}{x^2y + x^2z + y^2x + y^2z + z^2x + z^2y} > 4.$$

After expansion, this is equivalent to

$$x^3 + y^3 + z^3 + 6xyz > x^2y + x^2z + y^2x + y^2z + z^2x + z^2y.$$

This is true since $x^3 + y^3 + z^3 + 3xyz \geq x^2y + x^2z + y^2x + y^2z + z^2x + z^2y$ by Schur's inequality and $xyz > 0$.

Problem 2. (India MO 2020 Problem 4) Let $n \geq 2$ be an integer and let $1 < a_1 \leq a_2 \leq \dots \leq a_n$ be n real numbers such that $a_1 + a_2 + \dots + a_n = 2n$. Prove that

$$a_1a_2 \cdots a_{n-1} + a_1a_2 \cdots a_{n-2} + \dots + a_1a_2 + a_1 + 2 \leq a_1a_2 \cdots a_n.$$

Solution 1. We prove this by induction on n . For the base case $n = 2$, we have

$$\begin{aligned} & a_1 + 2 \leq a_1a_2 \\ \Leftrightarrow & a_1 + 2 \leq a_1(4 - a_1) \\ \Leftrightarrow & a_1^2 - 3a_1 + 2 \leq 0 \\ \Leftrightarrow & (a_1 - 1)(a_1 - 2) \leq 0. \end{aligned}$$

This is true since $1 < a_1 \leq a_2$ and $a_1 + a_2 = 4$ (which yields $1 < a_1 \leq 2$).

Assume the case n is established. Consider the case $n + 1$. Define $b = a_n + a_{n+1} - 2$. Since $a_{n+1} \geq \frac{2(n+1)}{n+1} = 2$, we have $b \geq a_n \geq a_{n-1}$. Therefore, we have

$$1 < a_1 \leq a_2 \leq \dots \leq a_{n-1} \leq b \quad \text{and} \quad a_1 + a_2 + \dots + a_{n-1} + b = 2(n+1) - 2 = 2n.$$

Using the inductive hypothesis, we obtain

$$a_1a_2 \cdots a_{n-1} + a_1a_2 \cdots a_{n-2} + \dots + a_1a_2 + a_1 + 2 \leq a_1a_2 \cdots a_{n-1}b.$$

Adding $a_1a_2 \cdots a_n$ to both sides, it remains to prove

$$a_1a_2 \cdots a_n + a_1a_2 \cdots a_{n-1}b \leq a_1a_2 \cdots a_{n+1}.$$

Now,

$$\begin{aligned} & a_1a_2 \cdots a_n + a_1a_2 \cdots a_{n-1}b \leq a_1a_2 \cdots a_{n+1} \\ \Leftrightarrow & a_n + (a_n + a_{n+1} - 2) \leq a_na_{n+1} \\ \Leftrightarrow & 0 \leq (a_n - 1)(a_{n+1} - 2). \end{aligned}$$

This holds because $a_n > 1$ and $a_{n+1} \geq 2$. Therefore, the inductive step is established, and the inequality is proved by induction. Equality holds when $a_1 = a_2 = \dots = a_n = 2$.

Solution 2. Since each a_i is larger than 1, we have $1 < a_1 < a_1 a_2 < \cdots < a_1 a_2 \cdots a_{n-1}$. Also, we have $a_1 \leq a_2 \leq a_3 \leq \cdots \leq a_n$. Applying Chebyshev's inequality to these two sequences, we obtain

$$\begin{aligned} & 1 \cdot a_1 + a_1 \cdot a_2 + a_1 a_2 \cdot a_3 + \cdots + a_1 a_2 \cdots a_{n-1} \cdot a_n \\ & \geq \frac{(1 + a_1 + a_1 a_2 + \cdots + a_1 a_2 \cdots a_{n-1})(a_1 + a_2 + \cdots + a_n)}{n}. \end{aligned}$$

Using $a_1 + a_2 + \cdots + a_n = 2n$, this becomes

$$a_1 + a_1 a_2 + a_1 a_2 a_3 + \cdots + a_1 a_2 \cdots a_n \geq 2(1 + a_1 + a_1 a_2 + \cdots + a_1 a_2 \cdots a_{n-1}).$$

This is the same as the desired inequality.

Remark. Another solution that we have seen in class is another formulation of solution 1. The important step in that solution is that we relax the condition and prove that inequality for any real numbers $a_1, a_2, \dots, a_n$ satisfying $1 < a_1 \leq a_2 \leq \cdots \leq a_n$ and $a_1 + a_2 + \cdots + a_n \geq 2n$ (not just equal to $2n$).

Problem 3. (India TST 2010) Let $A = (a_{jk})$ be a 10×10 array of positive real numbers such that the sum of numbers in each row as well as in each column is 1. Show that there exist $j < k$ and $\ell < m$ such that

$$a_{j\ell} a_{km} + a_{jm} a_{k\ell} \geq \frac{1}{50}.$$

Solution. Suppose on the contrary that $a_{j\ell} a_{km} + a_{jm} a_{k\ell} < \frac{1}{50}$ for any indices $j < k$ and $\ell < m$. We sum over all such inequalities. The product of each pair of entries not in the same row or in the same column appears exactly once. There are $\frac{9 \times 9 \times 100}{2 \times 2} = 2025$ inequalities in total. Therefore, we obtain

$$\sum_{\substack{j \neq k \\ \ell \neq m}} a_{j\ell} a_{km} < \frac{81}{2}.$$

Now, for a fixed $a_{j\ell}$, its coefficient in the above sum is

$$\sum_{\substack{k=1 \\ k \neq j}}^{10} \sum_{\substack{m=1 \\ m \neq \ell}}^{10} a_{km} = \sum_{\substack{k=1 \\ k \neq j}}^{10} (1 - a_{k\ell}) = 9 - (1 - a_{j\ell}) = 8 + a_{j\ell}.$$

Therefore, the inequality becomes $\sum_{j,\ell=1}^{10} a_{j\ell}(8 + a_{j\ell}) < 81$. Since the sum over all $a_{j\ell}$ is the sum of all rows, i.e. 10, this becomes

$$\sum_{j,\ell=1}^{10} a_{j\ell}^2 < 1.$$

However, by the power mean inequality, we have

$$\sum_{j,\ell=1}^{10} a_{j\ell}^2 \geq \frac{1}{100} \left(\sum_{j,\ell=1}^{10} a_{j\ell} \right)^2 = 1.$$

This is a contradiction. So one of the given inequalities must hold.

Problem 4. Given a convex polygon with $n \geq 3$ sides, let $a_1, a_2, \dots, a_n$ be its side lengths, and let p be its perimeter. Find the smallest constant C such that

$$\frac{a_1}{p-a_1} + \frac{a_2}{p-a_2} + \dots + \frac{a_n}{p-a_n} < C. \quad (1)$$

Solution 1. The answer is $C = 2$. Consider a convex polygon with two consecutive sides having length 1 and an extremely small included angle. The other vertices can be placed so that all other side lengths are close to 0. Assuming $a_1 = a_2 = 1$ and $a_3, \dots, a_n \rightarrow 0$, we have $p \rightarrow 2$, and

$$\frac{a_1}{p-a_1} + \frac{a_2}{p-a_2} + \dots + \frac{a_n}{p-a_n} \rightarrow 1 + 1 + 0 + 0 + \dots + 0 = 2.$$

This shows $C \geq 2$.

We now prove (1) with $C = 2$. By the triangle inequality, we have $a_1 < a_2 + a_3 + \dots + a_n = p - a_1$. So $2a_1 < p$, and hence $\frac{a_1}{p-a_1} < \frac{2a_1}{p}$. It follows that

$$\frac{a_1}{p-a_1} + \frac{a_2}{p-a_2} + \dots + \frac{a_n}{p-a_n} < \frac{2a_1}{p} + \frac{2a_2}{p} + \dots + \frac{2a_n}{p} = \frac{2p}{p} = 2.$$

This completes the proof.

Solution 2. This is an alternative proof to

$$\frac{a_1}{p-a_1} + \frac{a_2}{p-a_2} + \dots + \frac{a_n}{p-a_n} < 2.$$

Let $f(x) = \frac{x}{p-x}$. Then $f'(x) = \frac{p}{(p-x)^2}$ and $f''(x) = \frac{2p}{(p-x)^3} > 0$ for $0 \leq x < p$. Therefore, f is convex on $[0, p)$. WLOG assume $a_1 \geq a_2 \geq \dots \geq a_n$. By the triangle inequality as above, we have $a_1 < \frac{p}{2}$. So $\left(\frac{p}{2}, \frac{p}{2}, 0, 0, \dots, 0\right)$ majorizes $(a_1, a_2, \dots, a_n)$ and they are distinct sequences. By Karamata's inequality, we deduce

$$f\left(\frac{p}{2}\right) + f\left(\frac{p}{2}\right) + f(0) + f(0) + \dots + f(0) > f(a_1) + f(a_2) + \dots + f(a_n).$$

This is exactly the thing to prove as the left-hand side is 2.

Problem 5. Let $\{a_n\}$ be a sequence of real numbers such that $a_1 = 2$ and $a_{n+1} = \frac{2a_n^2 + a_n + 1}{a_n - 1}$ for

any positive integer n . Prove that $\sum_{k=1}^n \frac{1}{a_k} < \frac{2}{3}$ for any positive integer n .

Solution. Note that $a_{n+1} + 1 = \frac{2a_n^2 + a_n + 1}{a_n - 1} + 1 = \frac{2a_n^2 + 2a_n}{a_n - 1}$. So

$$\frac{2}{a_{n+1} + 1} = \frac{a_n - 1}{a_n(a_n + 1)} = -\frac{1}{a_n} + \frac{2}{a_n + 1}.$$

It follows that $\frac{1}{a_n} = 2 \left(\frac{1}{a_n + 1} - \frac{1}{a_{n+1} + 1} \right)$. Therefore, we have

$$\begin{aligned} \sum_{k=1}^n \frac{1}{a_k} &= 2 \left(\frac{1}{a_1 + 1} - \frac{1}{a_2 + 1} + \frac{1}{a_2 + 1} - \frac{1}{a_3 + 1} + \cdots + \frac{1}{a_n + 1} - \frac{1}{a_{n+1} + 1} \right) \\ &= 2 \left(\frac{1}{3} - \frac{1}{a_{n+1} + 1} \right) < \frac{2}{3}. \end{aligned}$$

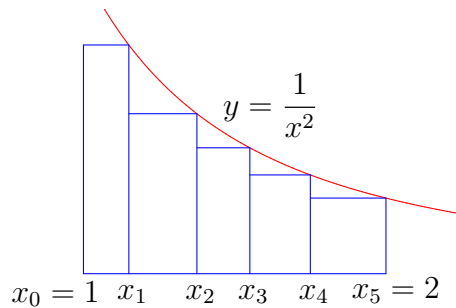
Problem 6. Considering all possible choices of the positive integer n and the positive real numbers $a_1, a_2, \dots, a_n$ with sum 1, find the smallest real number C such that

$$\frac{a_1}{(1 + a_1)^2} + \frac{a_2}{(1 + a_1 + a_2)^2} + \cdots + \frac{a_n}{(1 + a_1 + a_2 + \cdots + a_n)^2} < C.$$

Solution. Define $f(x) = \frac{1}{x^2}$ and let $x_j = 1 + a_1 + \cdots + a_j$ for $0 \leq j \leq n$ (so $x_0 = 1$). The graph of $y = f(x)$ is strictly decreasing on $(0, \infty)$. For each $j = 1, 2, \dots, n$, draw a rectangle on the xy -plane having vertices $(x_{j-1}, 0), (x_j, 0), (x_j, f(x_j)), (x_{j-1}, f(x_j))$. These rectangles lie under the graph $y = f(x)$ and are pairwise disjoint (except the edges). Therefore, the total area of all rectangles does not exceed the area under the graph on the interval $[1, 2]$. This total area is exactly

$$a_1 f(1 + a_1) + a_2 f(1 + a_1 + a_2) + \cdots + a_n f(1 + a_1 + a_2 + \cdots + a_n).$$

So $C \geq \int_1^2 \frac{1}{t^2} dt = \frac{1}{2}$. By the definition of integration, this bound is the best possible. Indeed, the total area of all rectangles approaches $\frac{1}{2}$ when $\max \{a_j\}$ approaches 0. For example, we may take $a_j = \frac{1}{n}$ for each j , and consider $n \rightarrow \infty$.



Problem 7. Let $0 < x < \pi$ and let n be a positive integer. Prove that

$$\frac{\sin x}{1} + \frac{\sin 2x}{2} + \cdots + \frac{\sin nx}{n} > 0.$$

Solution. We prove this by induction on n . The base case $n = 1$ is obvious since $\sin x > 0$ for $0 < x < \pi$.

Assume the inequality holds for some n . Consider

$$f(x) = \frac{\sin x}{1} + \frac{\sin 2x}{2} + \cdots + \frac{\sin (n+1)x}{n+1}.$$

Suppose on the contrary that $f(x) \leq 0$ for some $x \in (0, \pi)$. Note that $f(0) = f(\pi) = 0$. So we can choose the point of minimum of f on $[0, \pi]$ which lies in the open interval $(0, \pi)$. Let $t \in (0, \pi)$ be the point of minimum. Then $f(t) \leq 0$ and $f'(t) = 0$. Now,

$$\begin{aligned} f'(t) &= \cos t + \cos 2t + \cdots + \cos (n+1)t \\ &= \frac{1}{2 \sin \frac{t}{2}} \sum_{k=1}^{n+1} 2 \sin \frac{t}{2} \cos kt \\ &= \frac{1}{2 \sin \frac{t}{2}} \sum_{k=1}^{n+1} \left[\sin \left(kt + \frac{t}{2} \right) - \sin \left(kt - \frac{t}{2} \right) \right] \\ &= \frac{1}{2 \sin \frac{t}{2}} \left[\sin \left(nt + \frac{3t}{2} \right) - \sin \frac{t}{2} \right]. \end{aligned}$$

Since $f'(t) = 0$, this implies $\sin \left(nt + \frac{3t}{2} \right) = \sin \frac{t}{2}$. There are two possible cases.

If $nt + \frac{3t}{2} = \frac{t}{2} + 2m\pi$ for some $m \in \mathbb{Z}$, then $(n+1)t = 2m\pi$ so that $\sin (n+1)t = 0$. This yields

$$f(t) = \frac{\sin t}{1} + \frac{\sin 2t}{2} + \cdots + \frac{\sin nt}{n} > 0$$

by the inductive hypothesis.

If $nt + \frac{3t}{2} = (2m+1)\pi - \frac{t}{2}$ for some $m \in \mathbb{Z}$, then $(n+1)t = (2m+1)\pi - t$ so that $\sin (n+1)t > 0$ from $t \in (0, \pi)$. This yields

$$f(t) > \frac{\sin t}{1} + \frac{\sin 2t}{2} + \cdots + \frac{\sin nt}{n} > 0$$

by the inductive hypothesis.

In any case, this contradicts with the assumption $f(t) \leq 0$. So $f(x) > 0$ for any $x \in (0, \pi)$, and we are done by induction.

Problem 8. (China TST 2012 Test 2 Problem 2 modified) Prove that there exists a real number C satisfying the following property: for any integers $n \geq 2$ and $X \subseteq \{1, 2, \dots, n\}$, if $|X| \geq 2$, then there exist $x, y, z, w \in X$ (which can be the same) such that

$$0 < |xy - zw| < C\alpha^{-3},$$

where $\alpha = \frac{|X|}{n}$.

(Note: Weaker results with $C\alpha^{-3}$ replaced by $C\alpha^{-\beta}$ may be awarded points depending on the value of the constant $\beta > 3$.) :P

Solution 1. This is a solution to the original weaker version, where $\beta = 4$. We can take $C = 72$, for example. When $|X| = 2$, let x, y be the two distinct elements in X . Then

$$0 < |xx - yy| < n^2 < \frac{72}{16}n^4 = 72\alpha^{-4}.$$

Now we assume $|X| \geq 3$.

Claim. Let $S = \left\{ \frac{a}{b} : a, b \in X, a < b \right\}$. Then $|S| > \frac{\alpha^4 n^2}{36}$.

We first complete the proof by assuming this claim. Indeed, if $\frac{\alpha^4 n^2}{36} < 2$, then we can just pick any distinct $x, y \in X$ so that $0 < |xx - yy| < n^2 < 72\alpha^{-4}$. If $\frac{\alpha^4 n^2}{36} \geq 2$, then $|S| > 2$. Since every element in S lies between 0 and 1, there are two distinct elements $\frac{x}{z}$ and $\frac{w}{y}$ in S whose difference is at most $\frac{1}{|S| - 1}$ by the pigeonhole principle. Then

$$0 < |xy - zw| = yz \left| \frac{x}{z} - \frac{w}{y} \right| < \frac{n^2}{|S| - 1} < \frac{2n^2}{|S|} < 72\alpha^{-4}$$

as desired.

Now we are going to prove the claim. The result is trivial if $\frac{\alpha^4 n^2}{36} < 1$. So we may assume $\frac{\alpha^4 n^2}{36} \geq 1$, i.e. $\alpha^2 n \geq 6$. For each $d = 1, 2, \dots, n$, define

$$S_d = \{(a, b) : a, b \in X, a < b, \gcd(a, b) = d\}.$$

It is obvious that for any $(a, b) \in S_d$, the number $\frac{a}{b}$ is distinct. So it suffices to prove the existence of d such that $|S_d| > \frac{1}{36}\alpha^4 n^2$.

On the one hand, we have

$$\sum_{1 \leq d \leq n} |S_d| = \binom{|X|}{2} \geq \frac{|X|^2}{3} = \frac{\alpha^2 n^2}{3}. \quad (1)$$

On the other hand, let $M = \left\lceil \frac{3}{\alpha^2} \right\rceil < n$ (using $\alpha^2 n \geq 6$). We have

$$\sum_{1 \leq d \leq n} |S_d| = \sum_{1 \leq d \leq M} |S_d| + \sum_{M+1 \leq d \leq n} |S_d|. \quad (2)$$

Note that there are at most $\left\lfloor \frac{n}{d} \right\rfloor$ multiples of d in $\{1, 2, \dots, n\}$. So a trivial bound yields

$$|S_d| \leq \binom{\lfloor n/d \rfloor}{2} < \frac{n^2}{2d^2}.$$

Thus,

$$\sum_{M+1 \leq d \leq n} |S_d| < \frac{n^2}{2} \sum_{M+1 \leq d \leq n} \frac{1}{d^2} < \frac{n^2}{2} \sum_{M+1 \leq d \leq n} \frac{1}{d(d-1)} < \frac{n^2}{2M}. \quad (3)$$

Combining (1), (2) and (3), we find that

$$\sum_{1 \leq d \leq M} |S_d| > \frac{\alpha^2 n^2}{3} - \frac{n^2}{2M}.$$

By the pigeonhole principle, there exists d such that

$$|S_d| > \frac{\alpha^2 n^2}{3M} - \frac{n^2}{2M^2} > \frac{\alpha^2 n^2}{3 \left(\frac{3}{\alpha^2} + 1 \right)} - \frac{n^2}{2 \left(\frac{3}{\alpha^2} \right)^2} \geq \frac{\alpha^4 n^2}{12} - \frac{\alpha^4 n^2}{18} = \frac{\alpha^4 n^2}{36}$$

So we are done.

Solution 2. This is a solution to $\beta = 3$, where we can take $C = 100$. Let $|X| = k$. If $k^2 < 50n$, we simply consider distinct $x, y \in X$ and the difference $|xx - yy|$. Since this difference is less than n^2 and there are k different values of x^2 , by the pigeonhole principle, there must be a difference which is less than $\frac{n^2}{k-1} \leq \frac{2n^2}{k} < \frac{100n^3}{k^3} = 100\alpha^{-3}$. Now, we assume $k^2 \geq 50n$. Let $M = \left\lfloor \frac{5n}{k} \right\rfloor \leq \frac{k}{10}$.

Claim. There exists d with $1 \leq d \leq M$ and $a_1 < a_2 < \dots < a_m$ with $m \geq \frac{k^2}{10n}$ such that $a_i, a_i + d \in X$ for each $i = 1, 2, \dots, m$.

We first complete the proof by assuming this claim. By the pigeonhole principle, there exists an index i such that $a_i, a_i + d, a_{i+1}, a_{i+1} + d \in X$ and $a_{i+1} - a_i < \frac{n}{m-1} < \frac{2n}{m} < \frac{20n^2}{k^2}$. Then

$$0 < |a_i(a_{i+1} + d) - a_{i+1}(a_i + d)| = d(a_{i+1} - a_i) < M \cdot \frac{20n^2}{k^2} \leq \frac{100n^3}{k^3} = 100\alpha^{-3}.$$

Now we are going to prove the claim. Let $T_d = \{x : x, x + d \in X\}$ for each $d = 1, 2, \dots, M$. For $j = 1, 2, \dots, s$ where $s = \left\lceil \frac{n}{M} \right\rceil$, let c_j be the number of elements in X in the interval $((j-1)M, jM]$. By Jensen's theorem, we have

$$|T_1| + |T_2| + \dots + |T_M| \geq \binom{c_1}{2} + \binom{c_2}{2} + \dots + \binom{c_s}{2} \geq s \binom{k/s}{2} = \frac{k}{2} \left(\frac{k}{s} - 1 \right) > \frac{k}{2}$$

(using $s = \left\lceil \frac{n}{M} \right\rceil$ and $M = \left\lfloor \frac{5n}{k} \right\rfloor$). By the pigeonhole principle, there exists d with $1 \leq d \leq M$ such that $|T_d| > \frac{k}{2M} \geq \frac{k^2}{10n}$. This proves the claim, and so we are done.